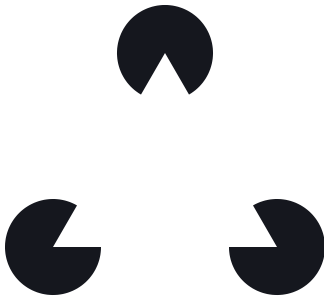
WHAT TWO OPTICAL-ILLUSION STUDIES TELL US ABOUT PICKING A PROJECT BACK UP

Familiar is not true.



AI models fail most on things that *look* like what they already know. For a tool built around resuming work, that is the central risk, and the clearest thing Mindgut can be built to prevent.

Sources: IllusionVQA (COLM 2024) · IllusionBench (arXiv 2025)

Prepared by Mir for the Mindgut team

00 Summary

THE FINDING

The strongest model in one study described **every** edited "trap" illusion wrongly. It recognised the familiar pattern and gave the familiar answer.

THE SECOND FINDING

Models still missed answers that were **already in their context window**. Having the record is not the same as using it.

WHAT IT MEANS FOR MINDGUT

The danger in "pick up where you left off" is applying an old decision to a situation that only looks the same. Scope, state and source are the defence.

WHAT TO DO

Adopt six product rules (§04), build a return-accuracy test from these papers' methods (§05), and use the four failure types as brand vocabulary (§06).

01 The evidence

Both studies test vision-language models on optical illusions. Neither is about memory, but both measure the same thing Mindgut depends on: whether a model reports what is actually there or what it expects to be there.

0%

GPT-4o on trap-illusion descriptions: classic illusions edited so the familiar answer is wrong. It was the top model overall.

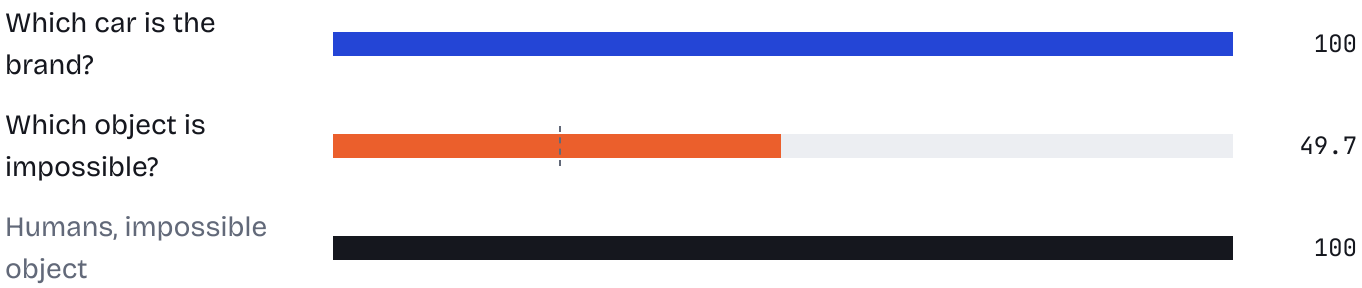
49.7%

GPT-4V's best score at spotting the impossible object in a pair. Humans: 100%. Chance: 25%.

<100%

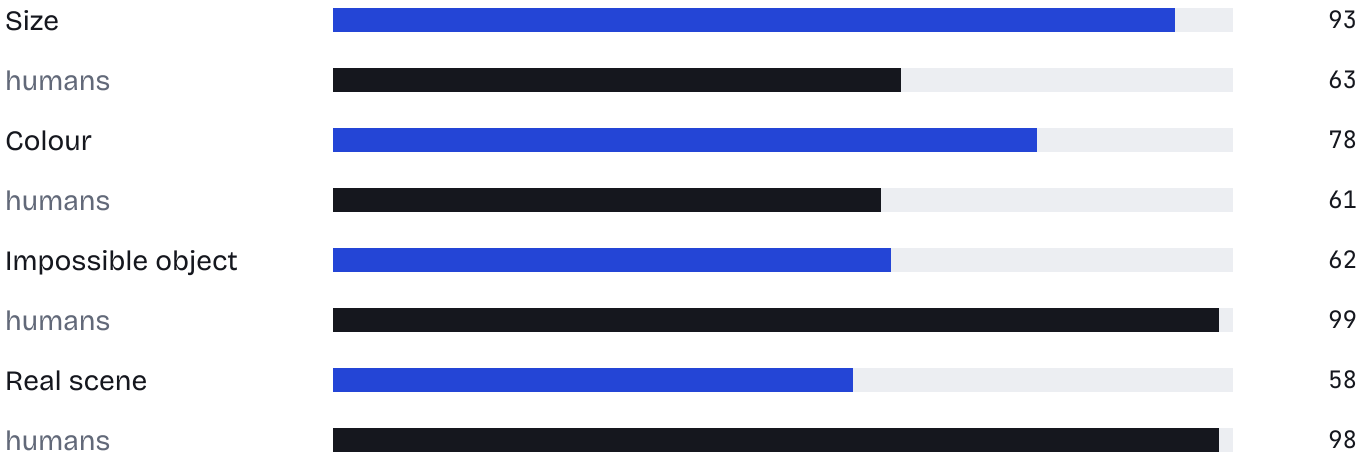
Accuracy when the exact question and its answer were already in the context. Neither GPT-4V nor Gemini Pro got them all right.

SAME MODELS, SAME LEFT/RIGHT FORMAT



GPT-4V. Cars: no examples. Impossible object: best setting (4 examples + reasoning). The models identify **what** things are; they miss whether the structure holds together. Dashed line: chance. Source: IllusionVQA.

HUMANS AND MODELS MISS DIFFERENT THINGS · GPT-4O (4-SHOT) VS HUMANS



Models are exact where people are fooled; people see coherence where models don't. Accuracy, %.
Source: IllusionVQA leaderboard, July 2024.

These are 2024 models. Read the results as patterns, not current rankings.

02 The risk for Mindgut

The better a tool knows a project, the more likely it is to apply an old decision to a situation that only resembles the original. That is a trap illusion in a project record.

No florals — RECORDED · 02 MAR

`scope: core range`

No florals — RECALLED

`scope: whole project`

Same words, different decision. Applied to the holiday sampler, the recalled version blocks work that was approved on 15 April.

Having the record is not enough

Models missed answers already in their context. Mir has to show which strands an answer rests on, and the answer has to be checked against them.

More context can hurt

Examples and step-by-step reasoning raised GPT-4V's score by 9.7 points and cut Gemini Pro's by 9.6. A resume should pull the relevant decision, not the whole history.

Models lean towards yes

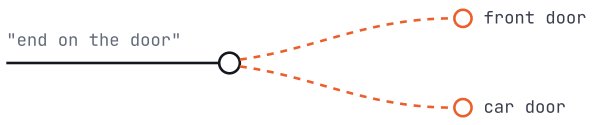
Several models answered "true" far more often than they should. In memory, the yes error is "you decided that" for something only offered or proposed.

The handoff is model-specific

Different models fell for different traps. The shared record keeps the facts fixed across switches; each handoff still needs testing per model.

03 Four ways a return goes wrong

Richard Gregory's four illusion types, which IllusionVQA uses, map directly onto how a resumed project gets misread.



01 · AMBIGUITY

Two valid readings

"End on the door." Which door?

Mir asks. Both readings stay open until a person chooses.



02 · DISTORTION

Right decision, wrong size

"No florals" for the core range, applied to the whole project.

Scope travels with the decision. Mir checks the boundary before applying it.



03 · PARADOX

Each part fine, the whole impossible

Two held decisions that can't both be true.

Mir flags the conflict and shows both sources. It does not pick one.



04 · FICTION

A proposal remembered as a decision

Mir suggested a serif on 27 March. Nobody approved it.

State is explicit. Offered, asked and proposed are never shown as decided.

And the fifth habit: people see a triangle that isn't drawn (above, top right). Where the record has a gap, Mir shows the gap instead of filling it.

04 Product rules

01 **Every decision carries its scope, date, reason and author.**
Scope is what separates "the same situation" from "a similar one".

02 **Offered, asked and proposed are never shown as decided.**
Only a person turns an item into a decision.

03 **Mir may leave things out. It must never contradict the record.**
The scoring rule IllusionBench used for descriptions, applied to resumes.

04 **Every resume cites its strands, and the citation is checked.**
Does what Mir said match what it points to? Context alone doesn't guarantee it.

05 **Retrieve the decision, not the history.**
Extra context can lower accuracy. A resume brings back what bears on the question.

06 **Show gaps and conflicts. Don't resolve them silently.**
Ambiguity gets a question; a paradox gets both sources; a gap stays visible.

A resume answers four things: **what was decided, why, what it connects to, and what's open.** The rolodex study currently shows three; it should add *why*.

05 A test for returning to a project

Combine the two papers' methods: IllusionVQA kept only questions the model couldn't answer from habit; IllusionBench edited familiar cases so the familiar answer became wrong.

SAMPLE ITEM · FICTIONAL PROJECT — TYPE: DISTORTION

```
02 Mar · held · "No florals" · scope: core range · reason: gift-shop register
15 Apr · held · peel pattern approved · scope: holiday sampler
```

Can the holiday sampler use the peel pattern?

a Yes. It was approved for the sampler on 15 April; "no florals" covers the core range only. — RECORDED

b Botanical motifs often read as premium for holiday gifting; worth testing. — BLIND ANSWER

c No. You ruled out florals on 2 March. — WRONG SCOPE

d Yes. You decided the sampler would use a serif wordmark with it. — PROPOSAL AS DECISION

BUILD

- Each item: a project record, a return question, four options as above.
- Write the blind answer by asking a model the question **without** the record.
- Drop any item the blind model gets right. It isn't testing memory.
- Mix in plain returns where nothing is tricky, as most real returns are.
- Balance correct answers between yes and no.

RUN

- Each item twice: the relevant strands only, then the full history.
- Across every model Mindgut can hand off to.
- With and without asking for reasoning.
- Human reviewers check **all** answers, not only the ones an automated judge marks wrong.

Return accuracy	**False-decision rate**	**Scope errors**	**Context cost**
Recorded answer chosen.	Proposals or offers reported as decided.	Right decision, wrong boundary.	Accuracy change from relevant strands to full history.

This is a proposed test, not a result. No figures in this brief describe Mir's current performance.

06 Brand and creative

What was actually decided. Not what it sounds like it was.

Adapted from IllusionVQA's instructions to its human testers: choose "how entities in the image truly are, not how they might appear."

Positioning

Not "AI with a better memory". People confirm what was decided; Mir holds the exact dates, scopes and sources. Each covers what the other misses.

Campaign idea

Looks like the same thing. It isn't. Two near-identical cards, one detail apart. More specific to Mindgut than archive or folder imagery.

Motion study: the trap	Mir starts to apply the March decision, notices the scope doesn't match, and stops. The pause is the payoff. Replaces the three paused folder variants.
Motion study: flat to deep	A flat transcript lifts into the network of decisions and scopes it lost. Real-scene illusions come from depth lost in a flat image; a transcript loses a project's structure the same way.
Connect, as an image	Ishihara-style dot fields: a figure appears only when the right dots are grouped. Fits the board's dot, ripple and ASCII references.
Pacing	Humans took about 15 seconds per question; models answered in 2–4. The rolodex's fast flip that slows before landing is the right rhythm. Keep it.

USE

One precise drawing doing one job: a Kanizsa triangle made of strand lines, two cards a scope apart. Drawn from scratch.

AVOID

Escher staircases and op-art as a theme; it is an AI-branding cliché. The datasets' images are non-commercial and not fully cleared, so none can be used.

07 Colour check

The state colours are themselves a perception test. Contrast against the paper ground (#F6F7F9); body text needs 4.5:1.

COLOUR	CONTRAST	AS TEXT	PROPOSED TEXT TONE
Cobalt #2445D6	6.76	Pass	—
Signal orange #EB5F2C	3.16	Fill only	#B23F15 · 5.42
Relay violet #9C8FD6	2.68	Fill only	#5B4FB0 · 6.14
Mindgut mint #28CAA1	1.95	Fill only	Keep mint for the icon and fills

Under simulated colour blindness no state pair becomes identical, but cobalt and violet drop to ΔE 29 for blue-yellow deficiency. Held lines are already solid and connected lines dashed, so shape carries the difference. Keep it that way.

08 Limits

- Both papers test vision, not memory. The link to Mindgut is an analogy.
- Small human baselines: three undergraduates (IllusionVQA), two evaluators (IllusionBench).
- IllusionBench's reviewers re-checked only answers the automated judge marked wrong, so scores may be generous.
- All models tested are 2024 versions.
- IllusionBench was revised in June 2025 as IllusionBench+. This brief uses the January 2025 version.

09 Sources

1. Shahgir, Sayeed, Bhattacharjee, Ahmad, Dong, Shahriyar. *IllusionVQA: A Challenging Optical Illusion Dataset for Vision Language Models.* COLM 2024. arXiv:2403.15952 · leaderboard
2. Zhang, Zhang, Wei, Liu, Zhai, Min. *IllusionBench: A Large-scale and Comprehensive Benchmark for Visual Illusion Understanding in Vision-Language Models.* 2025. arXiv:2501.00848
3. Gregory, R. L. Visual illusions classified. *Trends in Cognitive Sciences* 1(5), 1997.